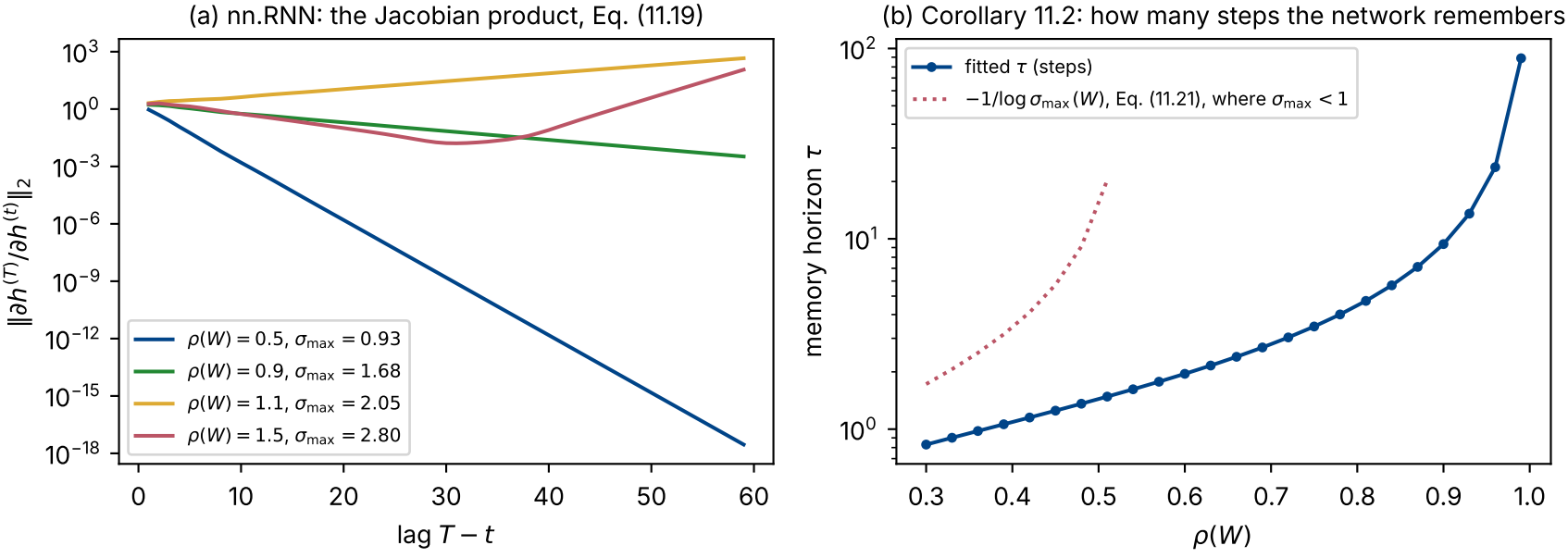
(a) nn.RNN: the Jacobian product, Eq. (11.19)
$\|\partial h^{(T)}/\partial h^{(t)}\|_2$
lag $T-t$
$\rho(W)=0.5$, $\sigma_{\max}=0.93$
$\rho(W)=0.9$, $\sigma_{\max}=1.68$
$\rho(W)=1.1$, $\sigma_{\max}=2.05$
$\rho(W)=1.5$, $\sigma_{\max}=2.80$
(b) Corollary 11.2: how many steps the network remembers
memory horizon τ
$\rho(W)$
fitted τ (steps)
$-1/\log\sigma_{\max}(W)$, Eq. (11.21), where $\sigma_{\max}<1$